

1. Derive  $\Omega_A$ ,  $\Omega_{MB}$ ,  $\eta_B$  and  $A_e$  for (a) a Hertz Dipole, (b) a rectangular aperture with a uniform illumination and (3) a circular aperture with a gaussian illumination.
2. For a Gaussian shaped main beam, the solid angle is given by

$$\Omega_{MB} = 1.133\theta_B^2$$

where  $\theta_B$  is the half power beam width (HPBW). **Prove this relation.** In the Submillimeter Array (SMA), each antenna has a diameter of 6 meter. Derive  $\theta_B$  and  $\Omega_{MB}$  at the observing frequency of 345 GHz, assuming a Gaussian illumination with a power taper of 20dB at the edge of the aperture. What is the total solid angle of the whole sky? How many beams are needed to fill up the whole sky? The surface of the antenna is not perfect and has an error of 20  $\mu\text{m}$  from the best-fit paraboloid, what is the shortest wavelength below which the antenna becomes unusable? What is the far-field distance at 345 GHz if the maximum departure  $\Delta$  at the edge of the reflector is required to be less than a tenth of a wavelength?

3. Suppose that a Gaussian-shaped source has an actual angular size  $\theta_s$  and actual peak temperature  $T_0$ . This source is measured with a Gaussian-shaped telescope (main) beam size  $\theta_B$ . The resulting peak temperature is  $T_{MB}$ . The flux density,  $S_\nu$ , integrated over the entire source, must be a fixed quantity, no matter what the size of the telescope beam. Use this argument to obtain following relation:

$$T_{MB} = T_0 \left( \frac{\theta_s^2}{\theta_B^2 + \theta_s^2} \right).$$

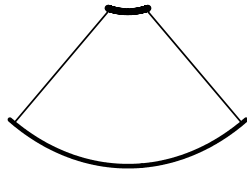
Show that when the source is small compared to the beam, the main beam brightness temperature  $T_{MB} \approx T_0(\theta_s/\theta_B)^2$ , and further the antenna temperature  $T_A \approx \eta_B T_0(\theta_s/\theta_B)^2$ . Suppose that a source has  $T_0 = 600$  K,  $\theta_s = 40''$ ,  $\theta_B = 8'$  and  $\eta_B = 0.6$ , what is  $T_A$ ?

4. At the point of closest approach, Saturn is 8.5 astronomical units (AU) from the earth. Saturn's radius is 60,000 km. Calculate the FWHP angular size, assuming a **disk-like shape**. Take the brightness temperature of Saturn as 130 K. What is the total flux density of Saturn at 8 mm?

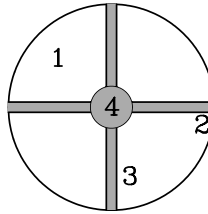
- a. Approximate the source by a Gaussian. If observed with the 100 m telescope, with a beam size of 25'', what is the main beam brightness temperature?
- b. If  $\eta_B$  is 0.7, calculate the peak antenna temperature.

5. In practice, many antenna designs require structures to hold the feed and/or subreflector in place, and these structures block part of the aperture. The situation is shown in the figure below, and can be analyzed by considering different parts of the structure separately, e.g. regions 1, 2, 3 and 4, and doing the Fourier Transform of each part.

Side view



Top view



Discuss how the blockage affects the power pattern, the beam shape and the aperture efficiency and why. If the area blocked is about 15% of the total area of the aperture, what is the aperture efficiency corrected for the blockage?